

Battery Guardian: Physical Consistency Model

1. Scope

The Guardian observes one sample every $\Delta t = 100$ ms:

$$x_k = (T_{\min,k}, T_{\text{avg},k}, T_{\max,k}, SoC_k).$$

The VSS/DBC interface provides minimum, average and maximum cell temperature as well as the battery state of charge. The DBC quantization is 0.5°C for temperatures and 0.5 percentage points for SoC; the message cycle time is 100 ms.

The Guardian does **not** estimate a detailed electro-thermal battery model. Instead, it checks a small set of closed-form physical consistency constraints. Each check compares an observed quantity (“IST”) against a state-dependent admissible limit (“SOLL”).

The model is intentionally simple and campaign-oriented. The numeric values below are model parameters for the demonstrator, not battery-cell qualification limits for a specific chemistry.

2. Configuration parameters

2.1 Sampling and transport

Parameter	Symbol	Proposed value	Unit	Meaning
Sample period	Δt	0.1	s	VSS/DBC sampling interval
Temperature quantization	q_T	0.5	$^\circ\text{C}$	DBC resolution
SoC quantization	q_{SoC}	0.5	pp	DBC resolution
Stale timeout	τ_{stale}	2.0	s	No fresh sample beyond this age
Watchdog interval	τ_{watchdog}	0.5	s	Stream freshness check interval

Here and below, pp means percentage points.

2.2 Absolute state limits

Parameter	Symbol	Proposed value	Unit
Absolute minimum battery temperature	$T_{\text{abs,min}}$	-30	°C
Absolute maximum battery temperature	$T_{\text{abs,max}}$	70	°C
Minimum SoC	SoC_{min}	0	%
Maximum SoC	SoC_{max}	100	%

2.3 Thermal state normalization

The temperature-dependent limits use a normalized thermal state

$$\theta(T) = \text{clip} \left(\frac{T - T_{\text{ref}}}{T_{\text{hot}} - T_{\text{ref}}}, 0, 1 \right).$$

Parameters:

Parameter	Symbol	Proposed value	Unit
Reference temperature	T_{ref}	20	°C
Hot-state temperature	T_{hot}	70	°C

Thus,

$$\theta(T) = \begin{cases} 0, & T \leq 20^\circ\text{C}, \\ \frac{T - 20}{50}, & 20 < T < 70^\circ\text{C}, \\ 1, & T \geq 70^\circ\text{C}. \end{cases}$$

The state variable is deliberately saturated. Temperatures outside the physical range are handled independently by the absolute-limit check.

2.4 Spatial consistency parameters

Parameter	Symbol	Cold value	Hot value	Unit
Maximum pack spread	S	$S_{\text{cold}} = 12$	$S_{\text{hot}} = 7$	$^{\circ}\text{C}$
Maximum hotspot deviation	H	$H_{\text{cold}} = 5$	$H_{\text{hot}} = 2$	$^{\circ}\text{C}$

2.5 Dynamic consistency parameters

Parameter	Symbol	Proposed value	Unit
Maximum heating rate, cold state	$R_{\uparrow, \text{cold}}$	8	$^{\circ}\text{C/s}$
Maximum heating rate, hot state	$R_{\uparrow, \text{hot}}$	5	$^{\circ}\text{C/s}$
Maximum cooling magnitude	$R_{\downarrow}$	6	$^{\circ}\text{C/s}$
SoC-to-heating coupling	K_{SoC}	0.25	$^{\circ}\text{C/pp}$
SoC-rate coupling cap	Q_{cap}	2	pp/s
Maximum SoC step	ΔSoC_{max}	0.5	pp/sample

The SoC coupling is intentionally weak and capped. It is optional; setting

$$K_{SoC} = 0$$

reduces the v1 model to a temperature-only dynamic bound without changing any other formula.

2.6 Stuck detection parameters

A raw rule such as “five identical samples” is not suitable because the temperature signals are quantized to 0.5°C and a healthy battery can legitimately remain unchanged over many 100-ms samples.

Instead, stuck detection uses a window and requires excitation elsewhere:

Parameter	Symbol	Proposed value	Meaning
Window length	N_{stuck}	10 samples	1 s observation window
Flatness tolerance	ϵ_{stuck}	0.25 °C	below half one temperature LSB
Temperature excitation	E_T	1.0 °C	another temperature channel changes sufficiently
SoC excitation	E_{SoC}	1.0 pp	SoC changes sufficiently

This detector is intended primarily for the controlled fault campaign. Without independent excitation or source metadata, a constant physical signal and a frozen sensor cannot always be distinguished.

3. Generic SOLL-IST formulation

For every invariant, define

$$r_k = \text{IST}_k - \text{SOLL}_k.$$

A positive residual indicates a violation:

$$r_k > 0 \implies \text{fault condition}$$

where appropriate, the residual is formed from the maximum of multiple one-sided violations.

This convention allows the Guardian to report the same generic evidence fields:

- **observed:** the actual measured quantity,
- **limit:** the state-dependent admissible limit,
- **residual:** `observed - limit`,
- **detection_class:** the violated invariant.

4. Physical consistency checks

4.1 Absolute temperature limit

Required condition:

$$T_{\text{abs,min}} \leq T_{\text{min},k} \leq T_{\text{max},k} \leq T_{\text{abs,max}}.$$

Residual:

$$r_{\text{abs},k} = \max(T_{\text{abs,min}} - T_{\text{min},k}, T_{\text{max},k} - T_{\text{abs,max}}).$$

Fault condition:

$$r_{\text{abs},k} > 0.$$

Guardian detection class:

PHYSICAL_TEMP_ABSOLUTE_LIMIT

4.2 Temperature ordering

Required condition:

$$T_{\text{min},k} \leq T_{\text{avg},k} \leq T_{\text{max},k}.$$

Residual:

$$r_{\text{ordering},k} = \max(T_{\text{min},k} - T_{\text{avg},k}, T_{\text{avg},k} - T_{\text{max},k}).$$

Fault condition:

$$r_{\text{ordering},k} > 0.$$

Guardian detection class:

PHYSICAL_TEMP_ORDERING

4.3 Temperature spread across the pack

The admissible maximum spread shrinks as the pack becomes hotter:

$$S(T) = S_{\text{cold}} - (S_{\text{cold}} - S_{\text{hot}}) \theta(T).$$

With the proposed parameters:

$$S(T) = 12 - 5 \theta(T)$$

in degrees Celsius.

Observed spread:

$$S_{\text{obs},k} = T_{\text{max},k} - T_{\text{min},k}.$$

Residual:

$$r_{\text{spread},k} = S_{\text{obs},k} - S(T_{\text{avg},k}).$$

Fault condition:

$$r_{\text{spread},k} > 0.$$

Guardian detection class:

PHYSICAL_TEMP_SPREAD

4.4 Hotspot deviation from the pack average

The hottest cell may only deviate by a state-dependent amount from the pack average:

$$H(T) = H_{\text{cold}} - (H_{\text{cold}} - H_{\text{hot}}) \theta(T).$$

With the proposed parameters:

$$H(T) = 5 - 3 \theta(T)$$

in degrees Celsius.

Observed hotspot deviation:

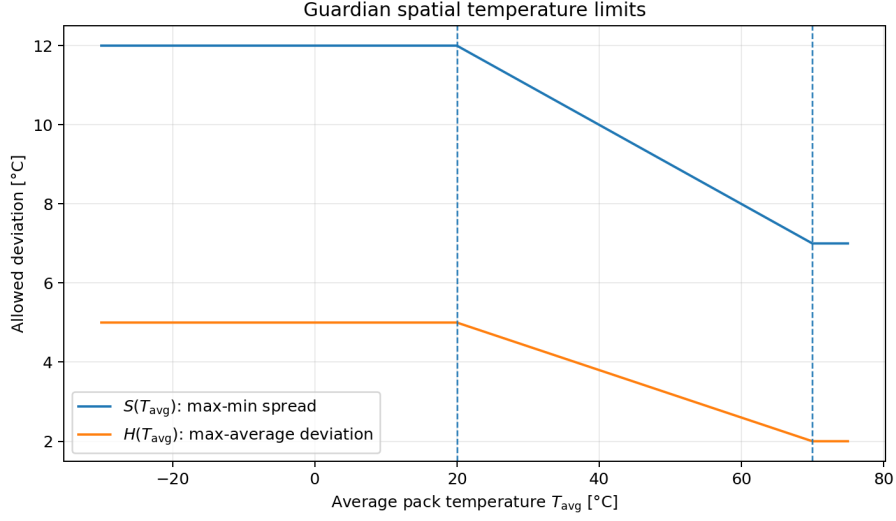


Figure 1: Temperature-dependent spatial limits

$$H_{\text{obs},k} = T_{\text{max},k} - T_{\text{avg},k}.$$

Residual:

$$r_{\text{hotspot},k} = H_{\text{obs},k} - H(T_{\text{avg},k}).$$

Fault condition:

$$r_{\text{hotspot},k} > 0.$$

Guardian detection class:

PHYSICAL_TEMP_HOTSPOT

The spread and hotspot checks are intentionally separate:

- **SPREAD** detects excessive non-uniformity over the complete pack.
- **HOTSPOT** detects the hottest cell diverging from the thermal bulk of the pack.

5. Dynamic temperature consistency

5.1 Measured temperature rate

For each temperature signal

$$i \in \{\text{min, avg, max}\},$$

compute

$$\dot{T}_{i,k} = \frac{T_{i,k} - T_{i,k-1}}{\Delta t}.$$

For the state-dependent limit, the **previous** average temperature is used:

$$T_{\text{state},k} = T_{\text{avg},k-1}.$$

This keeps the check causal and prevents the current anomalous sample from relaxing its own limit.

5.2 Base heating limit

$$R_{\uparrow,\text{base}}(T) = R_{\uparrow,\text{cold}} - (R_{\uparrow,\text{cold}} - R_{\uparrow,\text{hot}}) \theta(T).$$

With the proposed parameters:

$$\boxed{R_{\uparrow,\text{base}}(T) = 8 - 3\theta(T)} \quad [\text{°C/s}].$$

5.3 Optional SoC-dependent widening

The instantaneous SoC rate is

$$\dot{SoC}_k = \frac{SoC_k - SoC_{k-1}}{\Delta t}.$$

For the thermal coupling, use a capped absolute value

$$Q_k = \min(|\dot{SoC}_k|, Q_{\text{cap}}).$$

Then

$$\boxed{R_{\uparrow}(T, Q) = R_{\uparrow,\text{base}}(T) + K_{SoC}Q}$$

or numerically

$$R_{\uparrow}(T, Q) = 8 - 3\theta(T) + 0.25Q.$$

This is not intended as a detailed electro-thermal model. It only expresses the weak assumption that substantial charging/discharging may justify a somewhat larger positive temperature gradient.

The cap ensures that a corrupted SoC signal cannot arbitrarily relax the temperature plausibility bound.

5.4 Cooling limit

Cooling is bounded independently:

$$-R_{\downarrow} \leq \dot{T}_{i,k}$$

with

$$R_{\downarrow} = 6^{\circ}\text{C/s}.$$

5.5 Complete dynamic condition

For every $i \in \{\text{min}, \text{avg}, \text{max}\}$:

$$-R_{\downarrow} \leq \dot{T}_{i,k} \leq R_{\uparrow}(T_{\text{avg},k-1}, Q_k)$$

The one-sided residual can be written as

$$r_{\text{rate},i,k} = \max(\dot{T}_{i,k} - R_{\uparrow}, -\dot{T}_{i,k} - R_{\downarrow}).$$

The aggregate residual is

$$r_{\text{rate},k} = \max_i r_{\text{rate},i,k}.$$

Fault condition:

$$r_{\text{rate},k} > 0.$$

Guardian detection class:

PHYSICAL_TEMP_RATE

For a fixed 100-ms sample period, the positive per-sample bound is

$$\Delta T_{\uparrow, \max} = \Delta t \cdot R_{\uparrow}.$$

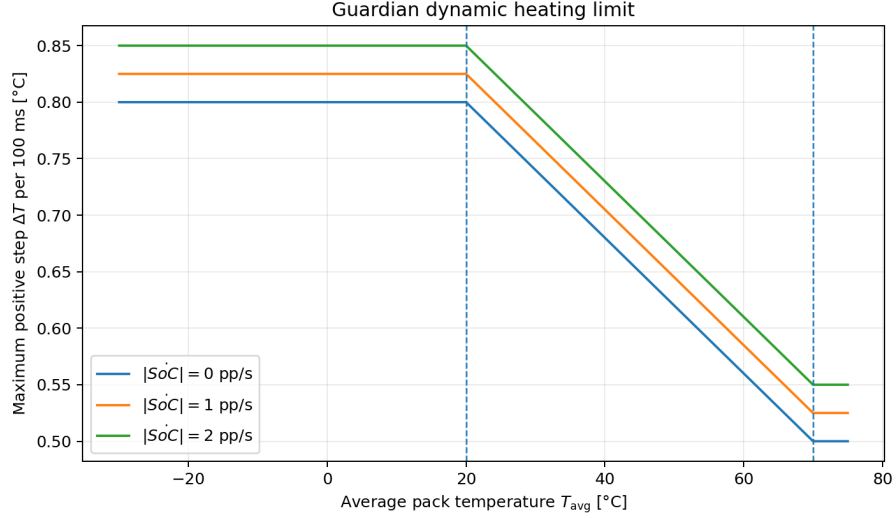


Figure 2: Maximum positive temperature step per sample

The proposed temperature-only base limit therefore decreases from

$$0.8^{\circ}\text{C/sample}$$

at the cold/reference state to

$$0.5^{\circ}\text{C/sample}$$

at the hot state.

The weak SoC coupling is shown separately:

6. State-of-charge consistency

6.1 SoC range

Required condition:

$$\text{SoC}_{\min} \leq \text{SoC}_k \leq \text{SoC}_{\max}$$

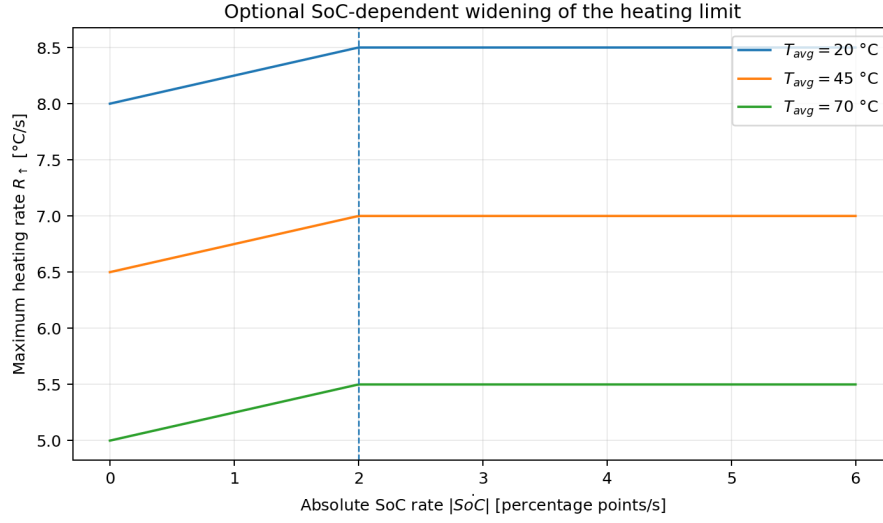


Figure 3: SoC-dependent heating-rate coupling

with

$$SoC_{\min} = 0, \quad SoC_{\max} = 100.$$

Residual:

$$r_{\text{soc-range},k} = \max(SoC_{\min} - SoC_k, SoC_k - SoC_{\max}).$$

Fault condition:

$$r_{\text{soc-range},k} > 0.$$

Guardian detection class:

PHYSICAL_SOC_RANGE

6.2 SoC step/rate

Because the DBC quantization is 0.5 percentage points, the v1 model uses a per-sample step bound:

$$\Delta SoC_k = SoC_k - SoC_{k-1}.$$

Required condition:

$$\boxed{|\Delta SoC_k| \leq \Delta SoC_{\max}}$$

with

$$\Delta SoC_{\max} = 0.5 \text{ pp/sample.}$$

Residual:

$$r_{\text{soc-rate},k} = |\Delta SoC_k| - \Delta SoC_{\max}.$$

Fault condition:

$$r_{\text{soc-rate},k} > 0.$$

Guardian detection class:

PHYSICAL_SOC_RATE

Equivalently, for $\Delta t = 0.1 \text{ s}$,

$$|\dot{SoC}_k| \leq 5 \text{ pp/s.}$$

The per-sample formulation is preferable here because it directly reflects the signal quantization.

7. Signal-stuck detection

For one signal x over the last N_{stuck} samples, define the observed range

$$A_x(k) = \max_{j=k-N+1, \dots, k} x_j - \min_{j=k-N+1, \dots, k} x_j.$$

The signal is locally flat if

$$A_x(k) \leq \epsilon_{\text{stuck}}.$$

For a temperature signal T_i , define external excitation as

$$E_i(k) = \left[\max_{j \neq i} A_{T_j}(k) \geq E_T \right] \vee [|SoC_k - SoC_{k-N+1}| \geq E_{SoC}].$$

Then

$$\text{stuck}_i(k) = \left[A_{T_i}(k) \leq \epsilon_{\text{stuck}} \right] \wedge E_i(k)$$

with proposed values

$$N_{\text{stuck}} = 10, \quad \epsilon_{\text{stuck}} = 0.25^\circ\text{C}, \quad E_T = 1.0^\circ\text{C}, \quad E_{SoC} = 1.0 \text{ pp.}$$

Guardian detection class:

SIGNAL_STUCK

This deliberately avoids classifying a thermally steady battery as faulty merely because the same quantized temperature value is observed repeatedly.

8. Stream freshness

Let t_{last} be the receive time of the most recent fresh battery event.

Stream age:

$$a(t) = t - t_{\text{last}}.$$

Required condition:

$$a(t) \leq \tau_{\text{stale}}.$$

Residual:

$$r_{\text{stale}}(t) = a(t) - \tau_{\text{stale}}.$$

Fault condition:

$$r_{\text{stale}}(t) > 0.$$

With

$$\tau_{\text{stale}} = 2.0 \text{ s.}$$

Guardian detection class:

STREAM_STALE

This is intentionally a **symptom** classification. In the current campaign, the following injected root causes may all lead to the same Guardian observation:

```

transport.delay  --\
transport.drop   ----+--> STREAM_STALE
source.dropout   --/

```

The Evidence Collector must therefore use campaign ground truth to distinguish the injected class.

9. Complete per-sample Guardian evaluation

For each new sample k :

1. Decode

$$T_{\min,k}, T_{\text{avg},k}, T_{\max,k}, SoC_k.$$

2. Check absolute temperature bounds:

$$r_{\text{abs},k}.$$

3. Check ordering:

$$r_{\text{ordering},k}.$$

4. Compute thermal state:

$$\theta_k = \theta(T_{\text{avg},k}).$$

5. Compute spatial limits:

$$S_k = S(T_{\text{avg},k}), \quad H_k = H(T_{\text{avg},k}).$$

6. Compare spatial IST values:

$$T_{\max,k} - T_{\min,k} \quad \text{and} \quad T_{\max,k} - T_{\text{avg},k}$$

against S_k and H_k .

7. If a previous sample exists, compute:

$$\dot{T}_{\min,k}, \dot{T}_{\text{avg},k}, \dot{T}_{\max,k}, \dot{SoC}_k.$$

8. Compute the dynamic heating limit from the previous thermal state:

$$R_{\uparrow,k} = R_{\uparrow}(T_{\text{avg},k-1}, Q_k).$$

9. Check temperature dynamics and SoC step.

10. Update the stuck-detection windows.

11. On every fault-state transition, emit a **GuardianEvidenceEvent**.

Separately, the watchdog evaluates stream freshness even when no new sample arrives.

10. Recommended Guardian evidence payload

For every state transition, the Guardian should expose enough data for the Evidence Collector to reproduce the SOLL-IST decision:

```
run_id: ...
detection_class: PHYSICAL_TEMP_SPREAD
fault_id: BatteryTempSpreadViolation
state: active
detected_at_ms: ...

context:
  temp_min: 42.0
  temp_avg: 47.0
  temp_max: 55.0
  soc: 68.5

  observed: 13.0
  limit: 9.3
  residual: 3.7

  model_variable: "temp_max-temp_min"
  state_variable: 47.0
```

For a dynamic violation, the context should additionally include the previous sample or derived rate:

```
context:
  signal: temp_max
  previous: 48.0
  current: 49.0
  delta_t_s: 0.1
  observed: 10.0      # degC/s
  limit: 6.5          # degC/s
  residual: 3.5
```

The Collector should not need to reimplement the physical model to decide whether the Guardian detected a violation. It should, however, receive **observed**, **limit**, and **residual** so that the detection is auditable.

11. Mapping to the current v1 fault campaign

Injected class	Primary Guardian observation
signal.stuck	SIGNAL_STUCK
signal.spike	PHYSICAL_TEMP_RATE

Injected class	Primary Guardian observation
signal.drift	PHYSICAL_TEMP_SPREAD, PHYSICAL_TEMP_HOTSPOT
signal.out_of_range	PHYSICAL_TEMP_ABSOLUTE_LIMIT
transport.delay	STREAM_STALE
transport.drop	STREAM_STALE
source.dropout	STREAM_STALE

A spike may legitimately violate more than one physical invariant. PHYSICAL_TEMP_RATE is the intended primary observation for the campaign scenario; the injected spike should therefore remain inside the absolute temperature range when possible.

Likewise, the drift scenario should remain within the per-sample rate limit and gradually violate the spread/hotspot constraints. This isolates the intended fault mechanism.

12. Compact parameter block

A direct implementation-oriented configuration could look like:

```
guardian_model:
    sample_period_ms: 100

    temperature:
        absolute_min_c: -30.0
        absolute_max_c: 70.0
        reference_c: 20.0
        hot_state_c: 70.0

    spread:
        cold_c: 12.0
        hot_c: 7.0

    hotspot:
        cold_c: 5.0
        hot_c: 2.0

    dynamics:
        heating_rate_c_per_s:
            cold: 8.0
            hot: 5.0
        cooling_rate_c_per_s: 6.0

    soc_coupling:
```

```
gain_c_per_pp: 0.25
rate_cap_pp_per_s: 2.0

soc:
  min_percent: 0.0
  max_percent: 100.0
  max_step_pp: 0.5

stuck:
  window_samples: 10
  flatness_epsilon_c: 0.25
  temperature_excitation_c: 1.0
  soc_excitation_pp: 1.0

stream:
  stale_timeout_ms: 2000
  watchdog_interval_ms: 500
```

For the simplest possible first implementation, the SoC-to-temperature coupling can be disabled with

```
gain_c_per_pp: 0.0
```

without changing the remainder of the Guardian model.